

```
In [4]: import pandas as pd
df = pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
print(df.head())
print(df.columns)
```

	Student ID	Student Name	Department	Age	Salary	Email
0	101	student1	IT	20.0	23075.0	student1@gmail.com
1	102	student2	ECE	18.0	35929.0	student2@gmail.com
2	103	STUDENT3	EEE	22.0	29718.0	student3@gmail.com
3	104	student4	MECH	20.0	21009.0	student4@gmail.com
4	105	Student5	CSE	19.0	47113.0	student5@gmail.com

```
Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary', 'Email'], dtype='object')
```

```
In [5]: #Rename columns
df.rename(columns={
    'Student ID': 'Student_ID',
    'Student Name ': 'Student_Name',
    ' Department': 'Department'
}, inplace=True)
print(df.columns)
```

```
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')
```

```
In [6]: #changing Data Types
#check Data Types
print(df.dtypes)
```

```
Student_ID      int64
Student_Name    object
Department      object
Age             float64
Salary          float64
Email           object
dtype: object
```

```
In [7]: #convert Student_ID to String
df['Student_ID'] = df['Student_ID'].astype(str)
#convert Age to Integer
#First handle missing values.
df['Age'] = df['Age'].fillna(0)
df['Age'] = df['Age'].astype(int)
#check
print(df.dtypes)
```

```
Student_ID      object
Student_Name    object
Department      object
Age             int64
Salary          float64
Email           object
dtype: object
```

```
In [8]: #Detecting Dupluicate Records
#find Duplicates
duplicates = df.duplicated()
print(duplicates)
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
10   False
11   False
12   False
13   False
14   False
15   False
16   False
17   False
18   False
19   False
20   False
21   False
22   False
23   False
24   False
25   False
26   False
27   False
28   False
29   False
30   False
31   False
32   False
33   False
34   False
35   False
36   False
37   False
38   False
39   False
40   False
41   False
42   False
43   False
44   False
45   False
46   False
47   False
48   False
49   False
50   False
51   False
dtype: bool
```

```
In [9]: #count Duplicate
print(df.duplicated().sum())
```

```
0
```

```
In [11]: #Dropping Duplicated Records
#before
print(df.shape)
#Remove Duplicates
df.drop_duplicates(inplace=True)
#After
print(df.shape)
```

(52, 6)

(52, 6)

```
In [14]: #string Operation
#string method work only on text columns
#remove Extra Space
df['Student_Name'] = df['Student_Name'].str.strip()
#convert to uppercase
df['Student_Name'] = df['Student_Name'].str.upper()
print(df['Student_Name'].head())
#convert to title case
df['Student_Name'] = df['Student_Name'].str.title()
print(df['Student_Name'].head())
```

0 STUDENT1

1 STUDENT2

2 STUDENT3

3 STUDENT4

4 STUDENT5

Name: Student_Name, dtype: object

0 Student1

1 Student2

2 Student3

3 Student4

4 Student5

Name: Student_Name, dtype: object

```
In [15]: #Replace Text
#Replace department name.
df['Department'] = df['Department'].replace({
    'CSE': 'Computer Science',
    'ECE': 'Electronics and Communication Engineering',
    'EEE': 'Electrical and Electronic Engineering',
    'MECH': 'Mechanical',
    'IT': 'Information Technology'
})
print(df['Department'].head())
```

0 Information Technology

1 Electronics and Communication Engineering

2 Electrical and Electronic Engineering

3 Mechanical

4 Computer Science

Name: Department, dtype: object

```
In [16]: #String Length
df['Name_Length'] = df['Student_Name'].str.len()
print(df[['Student_Name', 'Name_Length']].head())
```

	Student_Name	Name_Length
0	Student1	8
1	Student2	8
2	Student3	8
3	Student4	8
4	Student5	8

```
In [17]: #Extract First Letter
df['First_Letter'] = df['Student_Name'].str[0]
print(df[['Student_Name', 'First_Letter']].head())
```

	Student_Name	First_Letter
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S

```
In [ ]:
```